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FURNACE-X3 v3.0.0-soc-flagship Technical Product Report

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 31 October 2026 (2026-10-31 23:12 UTC+8 Beijing Time)

Abstract

FURNACE-X3 integrates an RV64IM firmware scheduler, native 512-bit burst DMA, hardware synchronization and interrupt aggregation with the preserved FURNACE-X1/X2 tensor, compression, cryptographic, vision and audio engines. Its modular furnace_x3_top supports autonomous dependency graphs while retaining reset-default legacy stream operation. Software-vs-Hardware numerical comparison passed against simulated RTL: independent Python models checked firmware execution and bit-exact DDR contents across five configurations and seeds 17, 89 and 2027. Recorded L1 results comprise 86 behavioral checks, ten detected behavioral mutants, eight synthesis-suite checks and fresh-source reproduction. Physical-board numerical parity is not established. [E119–E122] Native L1 RTL timing lint passed across 39 RTL files. L2 Intel Quartus Prime 17.1 TimeQuest measured WNS_setup = +5.672 ns at 10.000 ns for target 10AX115N2F45E1SG, with virtual pins except clk. This post-map analysis also recorded hold slack −0.203 ns and 20 failing hold paths. L3 physical Arria 10 FPGA HIL passed three deterministic bridge-connectivity probes, not X3 workload execution. These results do not establish post-fit timing closure or physical firmware/engine parity. [E124–E126]

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

The architecture extends rather than replaces the parent subsystem. All 26 parent modules remain project-local, preserving public X1/X2 interfaces and numerical behavior. A non-overlapped multicycle processor favors precise exceptions and simple transaction ownership over instruction throughput. Independent DMA, descriptor and frame resources permit concurrent work despite serialized CPU requests. Six memory clients share independent read/write arbitration, allowing one external read and one external write outstanding. DDR4 denotes the controller-facing AXI4 boundary; PHY, calibration and platform bridge integration remain external.

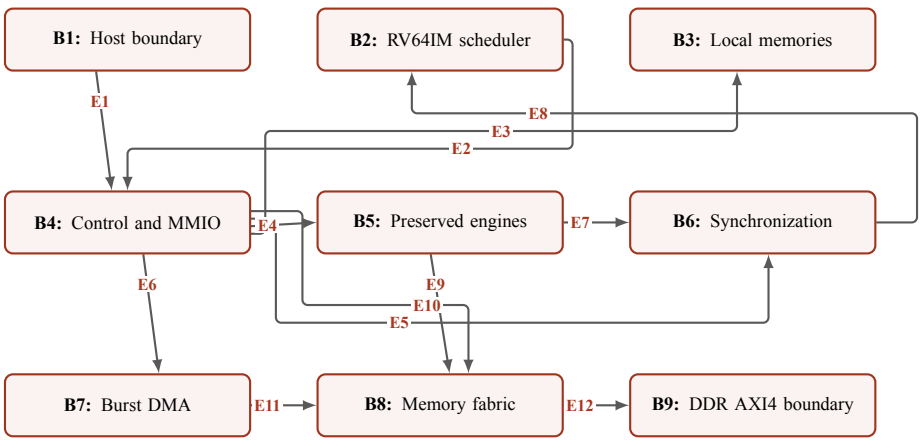


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Host boundary	AXI-Lite initialization, status and legacy external streams; host scheduling ends after RUN.
B2: RV64IM scheduler	control_cpu executes delivered firmware, resolves graph dependencies and handles precise machine-mode traps.
B3: Local memories	Separate 16 KiB instruction and data memories provide synchronous reads and byte-write enables.
B4: Control and MMIO	Fair host/CPU arbitration, address validation, register execution and sticky stream ownership.
B5: Preserved engines	legacy_soc retains authenticated compression/tensor processing, ISP, DSP and multimodal fusion.
B6: Synchronization	Sixteen semaphores, four mailbox FIFOs and interrupt aggregation account for engine and DMA completions.
B7: Burst DMA	Validated DDR byte copies use a 16-beat buffer, page splitting, exact tail strobes and completion posting.
B8: Memory fabric	Four legacy width adapters, CPU bridge and native DMA share independently arbitrated read/write channels.
B9: DDR AXI4 boundary	Scalar 512-bit master connects to the integrator's ordinary DDR memory controller.

Directed relationship	Meaning
E1: B1 to B4	Host requests and retained responses
E2: B2 to B4	Unified CPU transactions
E3: B4 to B3	Validated local-memory access
E4: B4 to B5	Parent CSRs and firmware stream slots
E5: B4 to B6	Atomic synchronization operations
E6: B4 to B7	DMA configuration and commands
E7: B5 to B6	Parent IRQ and frame completion
E8: B6 to B2	Machine-external interrupt
E9: B5 to B8	Four adapted legacy masters
E10: B4 to B8	CPU DDR transactions through bridge
E11: B7 to B8	AXI read/write bursts through client 5
E12: B8 to B9	Arbitrated 512-bit DDR transactions

Signal / group	Direction	Width	Contract

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n, irq	Clock/reset inputs; interrupt output	1 bit each	Single synchronous domain with asynchronous active-low reset. Public ready/valid, IRQ and response payloads are zero during reset. Interfaces enable after the first released-reset edge.
s_axil_*	AXI-Lite slave	Address 16; data 32; strobes 4; responses 2; protection 3 bits	Supports independent AW/W arrival and response backpressure. Parent registers remain at 0x0000–0x1fff; X3 registers occupy 0x2000–0x2fff. Host IMEM/DMEM apertures are 0x4000–0x7fff and 0x8000–0xbfff. Invalid operations return SLVERR and set protocol status.
m_axi_*	AXI4 master	Address 32; data 512; strobes 64; IDs 4; LEN 8; SIZE 3; BURST/ RESP 2; CACHE/ QOS/ REGION 4; PROT 3 bits	Handshake, LAST and LOCK signals are scalar. INCR bursts use SIZE=6, LOCK=0 and zero CACHE/ PROT/QOS/REGION. Physical IDs 0–3 identify legacy clients, 4 CPU and 5 DMA. Ownership persists through response completion; erroneous IDs reach the retained owner for explicit fault handling.
v_*, a_*, mm_*	Vision/audio inputs; multimodal output; reverse ready signals	Pixel 8; signed sample 16; IDs 16; result 64; error 8 bits	Valid/ready and LAST preserve parent packet semantics. FW_MODE=0 retains external streams. FW_MODE=1 disables external v_ready/a_ready/ mm_valid, substitutes registered firmware input slots and consumes results internally without depending on host mm_ready.
Internal CPU request/ response	Core master to memory/ control	Address/data 64; strobes 8; size 2 bits	One outstanding transaction; size encodes 1/2/4/8 bytes. Data is low-justified and held under stalls. Instruction fetches are four-byte reads. Unsupported upper address bits fault rather than truncate.

Parameter	Default	Meaning / supported values
MAX_TILE_BYTES	4096	1–65536 bytes; parent decoded/tile capacity.
RING_DEPTH	8	1–16 descriptor slots.
IN_WIDTH, IN_HEIGHT	8, 8	Each 1–64 pixels.
OUT_WIDTH, OUT_ HEIGHT	8, 8	Each 1–64 pixels.
FFT_N	16	8, 16, 32 or 64 samples.
FUSION_DEPTH	4	1–16 entries.
MAILBOX_DEPTH	4	1–16 words per mailbox, with explicit non-power-of-two wrap.
MAX_DMA_BYTES	1048576	1–1048576 bytes per command. Invalid parameter values terminate with PARAMETER ERROR.

III Behavior and corner cases

A Processor and cycle semantics

RV64IM includes W forms, sequential multiplication/division, machine Zicsr, precise traps, interrupts, MRET and WFI. Reset starts halted at PC 0x10000. x0 is immutable; W results sign-extend. Division implements architectural zero-divisor and signed-overflow results. M operations use 64 iterations, two normalization/sign stages and retirement. Fetch-response consumption precedes decode, execute and ordinary retirement by one, two and three edges. Handshakes use pre-edge VALID && READY; local requests execute one edge after capture and become response-consumable two edges after capture.

B Memory, ownership and errors

CPU IMEM is 0x10000–0x13fff, DMEM 0x20000–0x23fff and DDR 0x80000000–0xffffffff. IMEM is not CPU-writable; host local-memory writes while running fail. CPU MMIO requires aligned 32-bit accesses. FW_MODE can be set only while halted and parent traffic is quiescent, and remains set until global reset. Removing RUN waits for outstanding work to retire.

C DMA

Commands require 64-byte-aligned DDR endpoints, bounded nonoverlapping ranges and no address overflow. Zero length completes without traffic after address validation. Burst size is limited by remaining rounded-up beats, 16 beats and both 4 KiB page boundaries. Partial tails use exact byte strobes without read-modify-write. Success follows final B acknowledgement. Invalid commands return code 1; bus, ID or LAST faults return code 2 after necessary draining. Previously acknowledged writes may remain on failure.

D Synchronization

Semaphore counters saturate at 65535; hardware posts precede simultaneous software TAKE. DMA and internally consumed frame completions post semaphores 0 and 1. Overflow is explicit and sticky. Mailbox full simultaneous pop/push returns the old head and accepts replacement; empty pop/push has no bypass. IRQ update is (pending & ~ACK) | sources, so set wins clear. Sources include parent IRQ, completions, protocol errors, mailbox levels, overflow and core trap entry.

E Autonomous graph execution

The manifest identifies 1–64 DDR graph records of 64 bytes. Opcodes select DMA, an authenticated parent descriptor, multimodal frame or barrier. The scheduler supports fork/join dependencies, graph semaphores 2–15 and independent active resources. Descriptor retirement is polled; DMA/frame events use completion tokens and interrupt wait/resume. Sixteen-byte node statuses distinguish pending, active, success, failure and dependency failure. Final mailbox notification follows acknowledged status stores. Repeated graphs use core-only reset, preserving queue-wrap history.

F Tensor and authentication

The parent 128-byte descriptor ABI, AES-128/GCM ordering, AAD bytes 0–51 and LZ4/XXH32 validation are preserved. Failed authentication or framing releases no quarantined plaintext and commits no new weights. INT4×INT16 uses eight signed lanes and wrapping 64-bit accumulation. BF16/BF16 and INT4/BF16 extend BF16 into FP32 and round each product, balanced reduction level and accumulation separately to nearest-even, retaining gradual underflow, signed zero, infinity and canonical NaN 0x7fc00000.

G ISP, DSP and fusion

ISP uses floor align-corners Q8 coordinates, singleton coordinate zero, clamped bilinear interpolation with round-half-up and replicated spatial edges. Features are mean intensity and mean absolute Sobel/Laplacian responses. DSP resets FIR history per frame, applies [1,3,3,1]/8 with arithmetic floor shift, then radix-2 FFT using Q15 twiddles and signed-32 butterfly saturation. Spectral feature lanes saturate at 32767. Vision occupies low 64 activation bits and audio high 64. Arrival-ordered ID mismatch yields error 8 and zero data.

IV Verification methods and measured results

A Software-vs-Hardware numerical comparison

Method. Independent Python architectural/device and mathematical models compared firmware-driven simulated RTL, complete DDR images, records and guards.

Result. Five cells × three seeds passed within 86 behavioral checks, with zero asserted numerical mismatches. Seed-17 graph/descriptor/frame/mailbox counts were C0 9/18/9/9, C1 3/6/3/3, C2 7/14/7/7, C3 9/18/9/9 and C4 33/66/33/33. Physical-board numerical comparison is not evidenced.

Evidence: E119, E067, E088, E090, E076, E098

B Configuration and concurrency coverage

Method. Compiled Verilator C0–C4 runs used seeds 17, 89 and 2027; compact Icarus exercised actual firmware and engines.

Result. Configurations covered defaults; singleton geometry/depth and one-byte DMA; depth-3 upsampling with FFT32; downsampling with FFT64; and 64×64 geometry with depth16. Seed-17 overlap/multibeat counts were 13/36, 10/28, 12/36 and 43/132 for C0/C2/C3/C4. C1 recorded neither under its one-byte limit.

Evidence: E119, E076

C ISA, protocol and numerical boundaries

Method. Directed/random CPU signatures, AXI stalls/errors, synchronization races, crypto/LZ4 negatives, pipeline-edge assertions and legacy stream tests.

Result. Behavior suite passed, including three-seed ISA/numerical checks, singleton and non-power-of-two queues, parameter boundaries and error recovery.

Evidence: E119, E057, E065, E080, E091

D Defect sensitivity

Method. Ten disposable syntactically valid RTL mutants with corresponding passing original-source controls.

Result. All ten detected: W sign extension, division corner result, double TAKE, mailbox wrap, IRQ clear priority, DMA page split/tail/completion, authentication quarantine and fusion ordering. These injected defects do not imply their presence in production RTL.

Evidence: E120, E064

E Complete structural synthesis

Method. Seven complete-top Yosys synthesis runs covered X1, X2 and all five X3 cells, using retained packed-array lowering and behavioral controls. One additional check covered parameter diagnostics and source invariants.

Result. All eight synthesis-suite checks passed. Structural assertions required substantive connected hierarchy, zero latches, zero unresolved/black-box instances and zero division/modulo cells. Generic synthesis is not physical area or timing measurement.

Evidence: E121, E073, E080, E093, E096

F Portable reproduction

Method. Fresh directory containing declared sources only, regenerated firmware and fresh authenticated-weight/ISP/DSP/tensor/DDR smoke.

Result. One clean-source reproduction check passed without original generated executables or runtime metadata.

Evidence: E122, E083

G L1 RTL timing lint

Method. Native structural hazard analysis across 39 RTL files.

Result. Passed: zero combinational division/modulo, mixed sequential assignment, LUT clock-gate or loop-unroll hazards. This is the native L1.5 timing-lint tier, distinct from functional L1.

Evidence: E124

H L2 Intel Quartus Prime 17.1 TimeQuest STA

Method. Post-map analysis of clk_main at 10.000 ns for Arria 10 GX 10AX115N2F45E1SG, with VIRTUAL_PIN enabled on ports except clk.

Result. Passed: WNS_setup = +5.672 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 231.05 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E125

I L3 physical Arria 10 FPGA HIL

Method. Three consecutive PCIe hardware-bridge register probes on physical hardware.

Result. 3/3 deterministic passes, each reading register 57 as 178. Receipt scope is connectivity only; X3 bitstream/workload execution and physical numerical parity are not demonstrated.

Evidence: E126

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	9 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 51	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

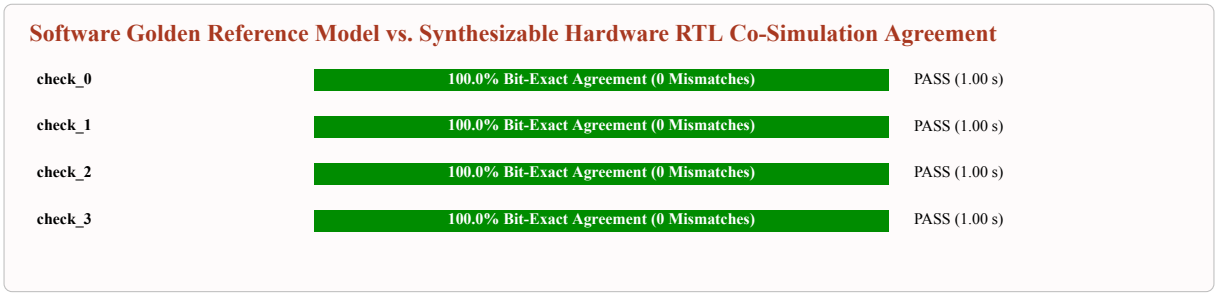


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +5.672 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_across_runs = True



Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

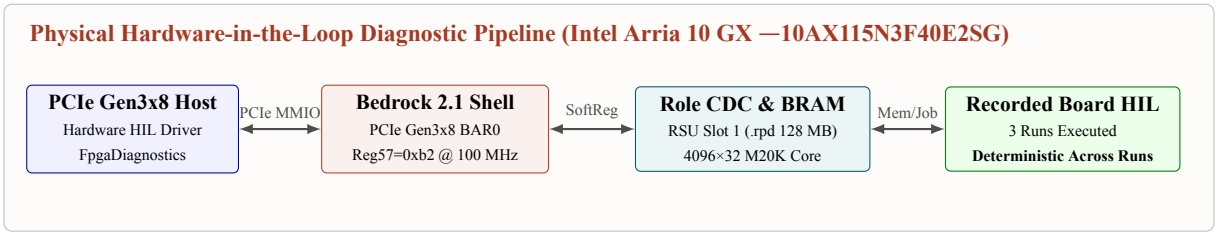


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

VI Integration and use

Use rtl/*.sv with furnace_x3_top and the documented parameters. Provide the external DDR controller/PHY and platform initialization path. Detailed maps and formats are in design/SPEC.md, design/ARCHITECTURE.md and firmware/README.md.

Generate the default executable with python3 firmware/build.py. For alternate geometry use python3 firmware/build.py --in-width 3 --in-height 2 --fft-n 32 --output firmware/c2.hex. Firmware geometry must match RTL.

While halted, initialize every readable instruction/data location, DDR graph/data/key material, parent ring and feature/status buffers. Load 32-bit image words through host IMEM aperture 0x4000 and the ten-word manifest through DMEM aperture 0x8000. DDR provisioning uses the external platform path, not AXI-Lite.

Quiesce legacy traffic, set FW_MODE at 0x2020, then write RUN at 0x2004. Perform no subsequent host scheduling writes. Observe DDR statuses and mailbox0 at 0x2600; drain notifications to preserve capacity.

For another graph, wait for halted status, update inputs and use core-only reset before RUN. Do not globally reset away ring, mailbox or fusion wrap history.

Portable dependencies are Python 3, Icarus/vvp, Verilator, make, g++ and Yosys. python3 tests/verify.py runs all L1 suites; independent selectors are --suite behavior, --suite defects, --suite synthesis and --suite portability. Native timing and hardware gates are separate.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 39 Verilog/SystemVerilog files; 7599 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/furnace_aes128_block.sv	145 total	145 code
rtl/furnace_aes128_gcm.sv	197 total	196 code
rtl/furnace_axi128_to512.sv	259 total	257 code
rtl/furnace_axi4_qos_xbar.sv	288 total	288 code
rtl/furnace_axi4_xbar.sv	282 total	282 code
rtl/furnace_bilinear_pipe.sv	68 total	67 code
rtl/furnace_command_ring.sv	44 total	44 code
rtl/furnace_csr.sv	187 total	187 code
rtl/furnace_dma128.sv	164 total	164 code
rtl/furnace_dma512.sv	257 total	253 code
rtl/furnace_dma_write_arb.sv	79 total	79 code
rtl/furnace_fft_radix2.sv	220 total	217 code
rtl/furnace_fir4_pipe.sv	56 total	56 code
rtl/furnace_fp32_add5.sv	124 total	124 code
rtl/furnace_fp32_mul4.sv	97 total	97 code

Production file	Physical lines	Code lines
rtl/furnace_fp32_round.sv	26 total	26 code
rtl/furnace_fusion_buffer.sv	86 total	86 code
rtl/furnace_ghash_mul.sv	35 total	35 code
rtl/furnace_hw_semaphores.sv	122 total	119 code
rtl/furnace_interrupt_aggregator.sv	103 total	100 code
rtl/furnace_lz4_frontend.sv	309 total	307 code
rtl/furnace_mailbox_fifo.sv	69 total	66 code
rtl/furnace_mailboxes.sv	120 total	117 code
rtl/furnace_rv64im_core.sv	468 total	461 code
rtl/furnace_spatial3x3_pipe.sv	58 total	58 code
rtl/furnace_tensor8.sv	139 total	139 code
rtl/furnace_x1_top.sv	506 total	502 code
rtl/furnace_x2_audio.sv	119 total	119 code
rtl/furnace_x2_csr.sv	267 total	267 code
rtl/furnace_x2_mm_control.sv	208 total	208 code
rtl/furnace_x2_top.sv	719 total	715 code
rtl/furnace_x2_vision.sv	280 total	280 code
rtl/furnace_x3_axi512_arb.sv	194 total	190 code
rtl/furnace_x3_control.sv	421 total	417 code
rtl/furnace_x3_cpu_axi_bridge.sv	161 total	159 code
rtl/furnace_x3_local_memory.sv	84 total	83 code
rtl/furnace_x3_mmio_arb.sv	135 total	134 code
rtl/furnace_x3_top.sv	438 total	436 code
rtl/furnace_xxh32.sv	119 total	119 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E057	tests/cpu_isa_checks.py
E064	tests/defect_checks.py
E065	tests/dma_checks.py
E067	tests/firmware_reference_checks.py
E073	tests/lower_packed.py
E076	tests/matrix_checks.py
E080	tests/parameter_checks.py
E083	tests/portability_checks.py
E088	tests/rv64_reference.py
E090	tests/soc_reference.py
E091	tests/synchronization_checks.py
E093	tests/synthesis_checks.py
E096	tests/verify.py
E098	tests/x3_smoke_tb.sv
E119	.vulcan/engineering/run-0001/attempt-0021/stdout.log
E120	.vulcan/engineering/run-0001/attempt-0022/stdout.log
E121	.vulcan/engineering/run-0001/attempt-0023/stdout.log
E122	.vulcan/engineering/run-0001/attempt-0024/stdout.log
E124	.vulcan/engineering/run-0001/rtl-timing-lint.json
E125	.vulcan/engineering/run-0001/quartus-sta.json
E126	.vulcan/engineering/run-0001/board-hil.json

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).